Vinith Ayapilla

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SUMMARY

Analyst with an MBA who turns sparse, messy, multi-source data into decisions people can act on. Currently translating international assessment data into policy recommendations for education ministries across participating countries, and independently authored a policy analysis cross-referencing IEA achievement data with World Bank and labor-market data to explain why a country's education spending wasn't converting into outcomes. Earlier, spent two and a half years on healthcare and pharmaceutical commercial data at ZS Associates, including patient-side analytics and GTM strategy across three specialty drug launches. Building and evaluating AI systems hands-on, so I have concrete views on where the technology helps and where the claims outrun the evidence. Grew up in India, which shapes how I think about which interventions actually reach people.

SKILLS

Cost-Effectiveness & Back-of-Envelope Modeling/Quick Math, Multi-Source Evidence Synthesis, Policy Analysis & Recommendations, Quantitative Research Design, Healthcare & Pharmaceutical Data Analysis, LMIC & Cross-National Data, Stakeholder & Expert Interviewing, Reasoning Under Uncertainty, Written Synthesis for Senior Decision-Makers, AI/LLM Systems & Evaluation, SQL, Python, Statistical Reporting, Interactive Dashboards

EXPERIENCE

IEA TIMSS & PIRLS *Chestnut Hill, MA*

Technical Dev/Consultant - AI Evaluation & Measurement May 2026 – Current

- Translate cross-national assessment results into structured, actionable insights for education ministries and policymakers across participating countries, many of them low- and middle-income
- Own the research program applying LLM comparative judgment and psychometric calibration to item-difficulty estimation, including explicit treatment of judge reliability, extrapolation limits, and where estimates should not be trusted (two paper submissions; see below)
- Build interactive data visualizations enabling global stakeholders to explore and interrogate complex educational datasets directly
- Migrated internal AI tooling to open-source models hosted on-premise to comply with EU GDPR data-residency requirements

AI Engineer Sep 2025 – May 2026

- Designed and implemented a scoring framework evaluating 10,000+ open-ended model outputs against ground truth, fine-tuning open-source weights to improve reasoning consistency at scale
- Built a Human-in-the-Loop verification layer to catch data-quality and sourcing errors across a multi-model ingestion pipeline (OpenAI, Gemini) before they propagated into fine-tuning data
- Designed a model cascade routing across quantized open-weight models with API fallback under fixed hardware constraints, trading capability against cost and reliability per request type
- Led the migration from proprietary OpenAI APIs to self-hosted open-source LLMs, cutting per-question costs from ~$60 to near-zero operational compute

Boston College *Chestnut Hill, MA*

Sr. Data Analyst | Awards/Grants Intake Team | Graduate Assistant Nov 2024 – Sep 2025

- Owned the data lifecycle for university research grants, serving as primary data advisor to the Vice-Provost; built automated reporting workflows and a dashboard suite that cut reporting time by 75%

ZS Associates *NCR, India*

Associate | Business Technology Jan 2022 – Jun 2024

- Architected a real-time geo-analytics tool to optimize territory planning and resource allocation, processing large-scale demographic and market data to identify high-potential geographic regions and market capture opportunities - directly driving $500K in new annual revenue.
- Diagnosed the root cause of a flawed, unoptimized data pipeline producing unreliable client-facing data; built a proof-of-concept to validate a re-engineered approach (SQL, Python, Shell) that cut manual QC intervention by 50% and restored data integrity.
- Managed commercial data strategy for 3 major product launches, coordinating cross-functional teams (Sales, Marketing, IT) to align stakeholders and ensure go-to-market readiness for assets representing 80% of client revenue.

PROJECTS & RESEARCH

Kosovo Education Policy Analysis: The Enthusiasm–Achievement Paradox *July 2026*

Authored an independent multi-pillar policy analysis cross-referencing IEA PIRLS/TIMSS achievement data with World Bank and Kosovo labor-market data to explain the gap between high student engagement and low cognitive-skill conversion which spans across socioeconomic, schooling infrastructure, curriculum, and brain-drain dimensions and proposes a policy roadmap.

Auditing LLM Comparative Judges Against an External Latent Criterion *- under review* *NeurIPS 2026*

Audited an LLM pairwise judge across 12,804 item pairs presented in both orders against withheld operational IRT difficulty. Found 11.06% order-instability concentrated among near-ties, accounting for ~95% of observed intransitivity; showed that filtering unstable judgments produced near-perfect transitivity without improving criterion validity - a more internally consistent judge is not a more valid one.

Designing Efficient LLM Comparative Judgment Networks for Item Difficulty Estimation *NCME 2027 submission*

Evaluated how anchor information and comparison-network design affect LLM-based item-difficulty estimation across 1,351 Bradley-Terry fits over 28 network designs. Showed that explicit calibrated anchor information improves scale recovery without improving rank recovery, and that sparse networks (22% of the full comparison budget) retain 98% of rank correlation but produce badly overdispersed estimates, i.e. efficient ranking designs shouldn't be mistaken for calibrated measurement.

AI Safety Evaluation - Uncensored vs. Baseline Model Benchmarking *August 2026*

Ran a local reimplementation of mini Vending-Bench Mini alongside AgentHarm, StrongREJECT and WMDP to compare an uncensored open-weight model (Gemma-4 26B) against its stock baseline, using Gemini as judge under hardware-constraints. The abliterated model performed worse - went out of business in all trials compared to the original which went out of business only about ~50% of the time and this was expected due to manual weight surgery on the refusal vector.

EDUCATION

Boston College | Carroll Graduate School of Management *Boston, May 2026*
MBA | STEM Track | Strategy | Data | GMAT 740 | Recipient of Dean's Scholarship | CGPA – 3.64/4.00

Shiv Nadar University *India, May 2022*
Bachelor of Technology in Electrical & Electronics Engineering | Minor in Big Data Analytics